

INSTITUTIONAL BRIEFING · ARTIFICIAL INTELLIGENCE

Retrieval-Augmented Generation (RAG)

Grounding large language models in real, verifiable knowledge

Foundations · Evidence · Evolution · Production



Six parts, from concept to production

01**Foundations**

Why LLMs need retrieval, what RAG is, where it is used

02**How it works**

Indexing, chunking, embeddings, retrieval and generation

03**Evidence**

Published results that quantify the gains

04**Evolution of RAG**

Generations, dimensions of progress, paradigms

05**Advanced techniques**

Toolbox; RAG vs fine-tuning vs long context

06**Production**

Evaluation, security, tooling, failure points

SECTION

01

Foundations

Why do language models need an external memory?



Why an LLM alone is not enough

An LLM is a brilliant reasoner...

...but an unreliable library.

Its knowledge is stored implicitly in its weights (parametric memory): frozen, opaque and impossible to cite.



Knowledge cutoff

Anything that happened after training is unknown to the model.



Hallucinations

When knowledge is missing, the model can produce fluent but unsupported answers.



No private data

Internal documents, wikis, tickets and databases were never in the training set.



No traceability

Answers come without sources, so users cannot verify them.

What is Retrieval-Augmented Generation?

RAG retrieves relevant passages from an external knowledge source at query time and injects them into the prompt, so the LLM generates an answer **grounded in that evidence**.



Closed-book exam

LLM alone

Answers from memory only (model weights). May be outdated or invented.

Parametric memory only



Open-book exam

LLM + RAG

Looks up the right pages first, then answers — and can cite them.

Parametric + non-parametric memory

ORIGIN

Lewis et al., 2020

“Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks” — NeurIPS 2020 (Facebook AI Research, UCL, NYU).

Combines parametric memory (BART) with non-parametric memory (a DPR index over Wikipedia).

Where RAG creates value



Customer support

Assistants answer from help-center articles, policies and past tickets, with links to sources.



Enterprise knowledge

One question-answering layer over wikis, intranets, drives and chat history.



Legal & compliance

Search contracts, regulations and case files; every statement traceable to a clause.



Healthcare & research

Ground answers in guidelines and scientific literature instead of model memory.



Software engineering

Code assistants retrieve relevant files, APIs and internal documentation.



Finance & analysis

Question reports, filings and market research that change every quarter.

Common pattern: a large, changing or private corpus + a need for answers users can verify.

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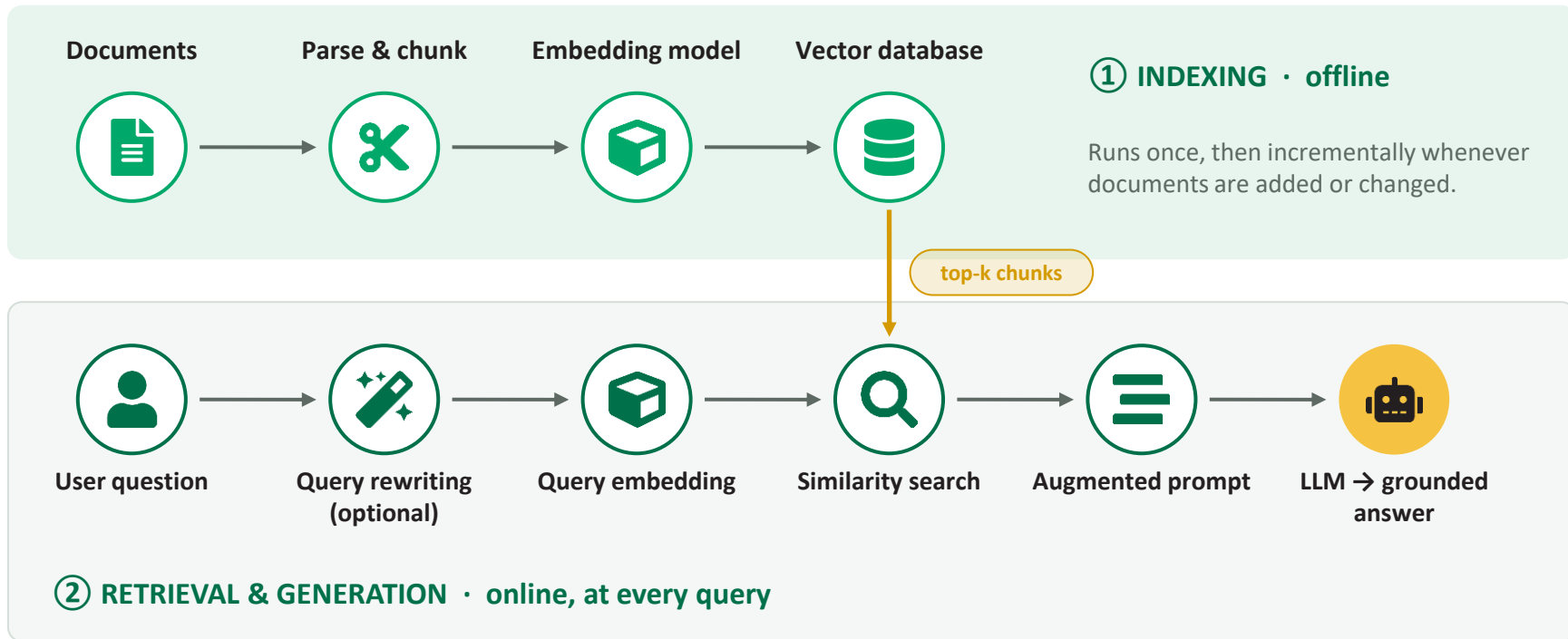
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How it works

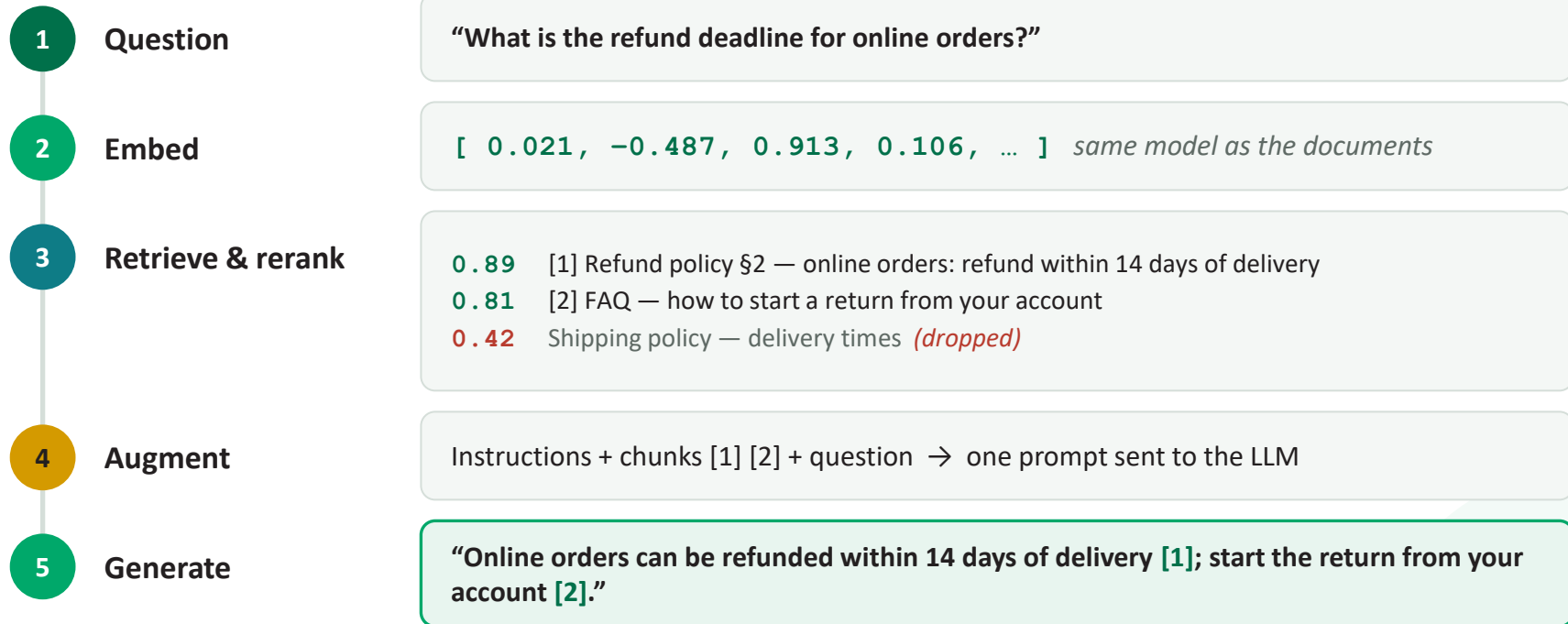
What happens between a question and a grounded answer?



Two pipelines: indexing and querying

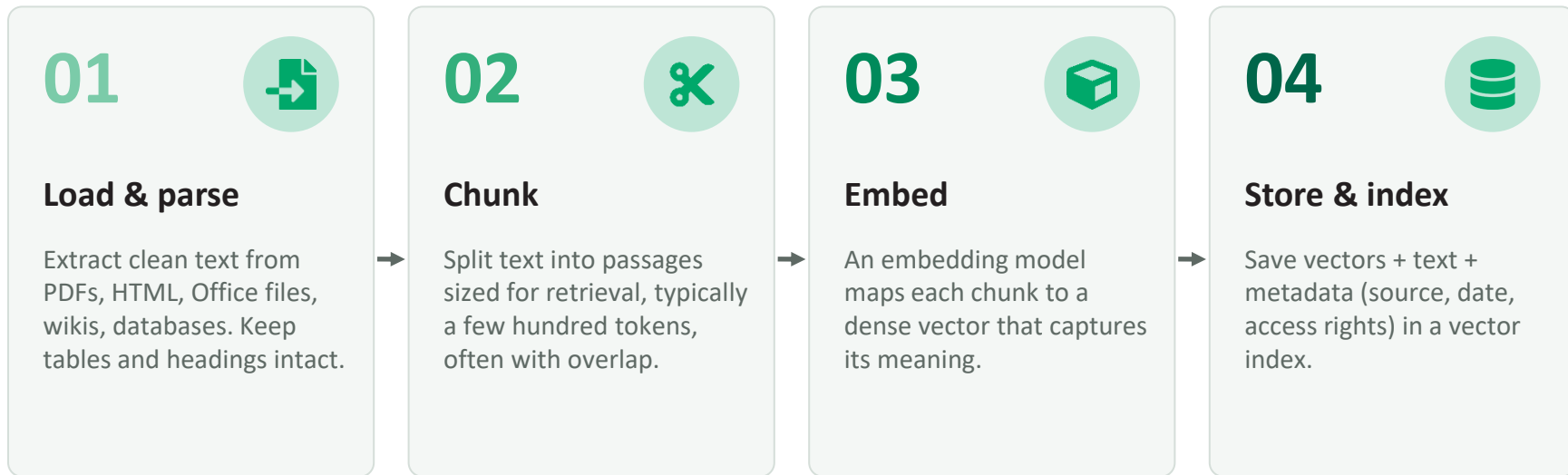


RAG in action: one question, end to end



Illustrative example — vectors, scores and documents are simplified.

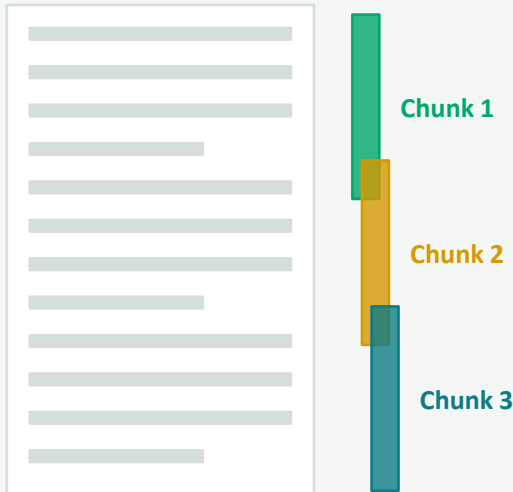
Turning documents into searchable knowledge



Key rule: the same embedding model must be used for documents and for queries — otherwise the vectors are not comparable.

The most underestimated design decision

Fixed-size chunks with overlap



Overlapping zones keep sentences that cross a boundary retrievable.

Common strategies

1

Fixed-size + overlap

Simple and predictable; may cut ideas mid-sentence.

2

Recursive splitting

Splits on paragraphs, then sentences, then words until the size fits.

3

Semantic chunking

Starts a new chunk where the embedding similarity between sentences drops.

4

Structure-aware

Follows headings, sections, tables, code blocks or Markdown structure.

Too small

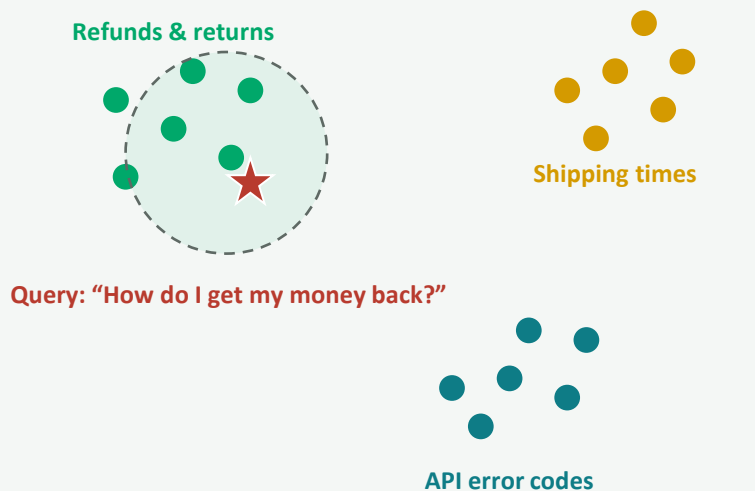
Precise match, but context is lost.

Too large

Rich context, but diluted, noisy vectors.

Embeddings: searching by meaning, not by words

Embedding space (2-D projection, illustrative)



No shared keyword with "refund", yet its vector lands next to refund passages.

Similarity measure

$$\cos(\mathbf{q}, \mathbf{d}) = (\mathbf{q} \cdot \mathbf{d}) / (\|\mathbf{q}\| \|\mathbf{d}\|)$$

Score close to 1 = same meaning. Dot product and Euclidean distance are also common.

Approximate Nearest Neighbour indexes

HNSW

Multi-layer proximity graph; fast, high recall

IVF

Clusters vectors; searches only nearby clusters

PQ

Compresses vectors to cut memory usage

ANN trades a tiny loss of exactness for millisecond search over millions of vectors.

Sparse, dense, hybrid — then rerank



Sparse · BM25

Keyword matching with term-frequency weighting.

Best for exact terms, product codes, names, rare words.



Dense · Embeddings

Vector similarity on meaning.

Best for paraphrases, synonyms, natural-language questions.



Hybrid

Runs both and fuses the ranked lists, e.g. with Reciprocal Rank Fusion.

$$RRF(d) = \sum 1 / (k + \text{rank}_i(d))$$

RERANKING

Two-stage retrieval: fast recall first, precise scoring second.

Top-50 / 100 candidates

Bi-encoder or BM25 · fast



Cross-encoder rescoring

Reads query + passage together · accurate



Top-3 to 10

Sent to the LLM

Building the augmented prompt

SYSTEM

Answer using ONLY the context below. Cite sources as [n]. If the answer is not in the context, say you don't know.

CONTEXT

[1] Refund policy, \$2 – Online orders are refunded within 14 days of delivery...
[2] FAQ – Returns are started from your account...

QUESTION

What is the refund deadline for online orders?

Illustrative template



Require citations

Each claim is linked to a retrieved chunk the user can check.



Allow abstention

“I don't know” is better than a confident hallucination.



Order matters

Place the strongest evidence first: models under-use the middle of long contexts (see Evidence).



Keep context focused

Fewer, more relevant chunks usually beat many noisy ones.

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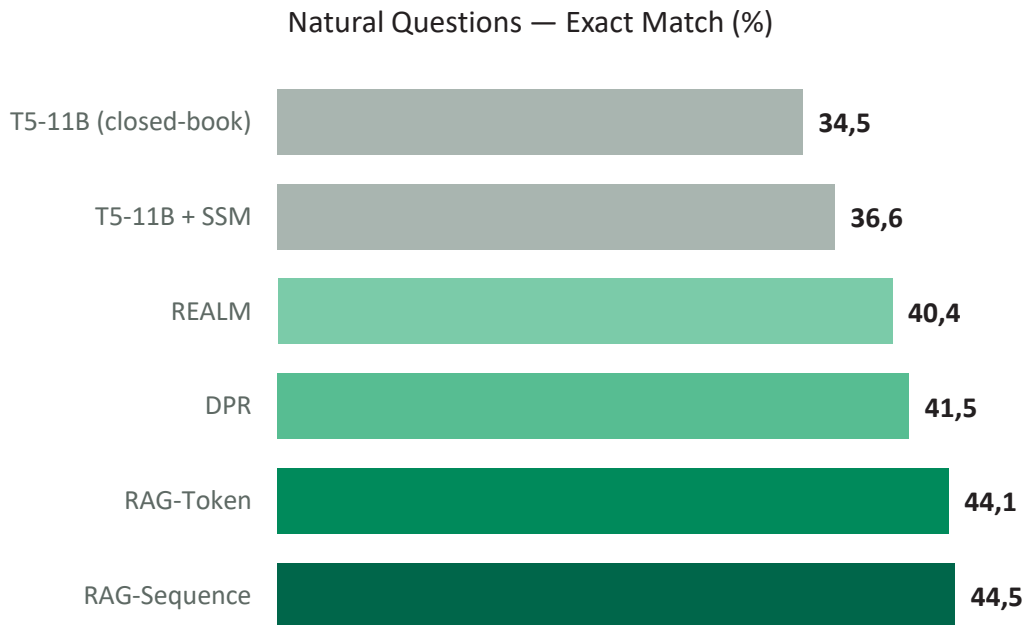
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Evidence

Does retrieval really improve answers?



Retrieval beats sheer model size



626M

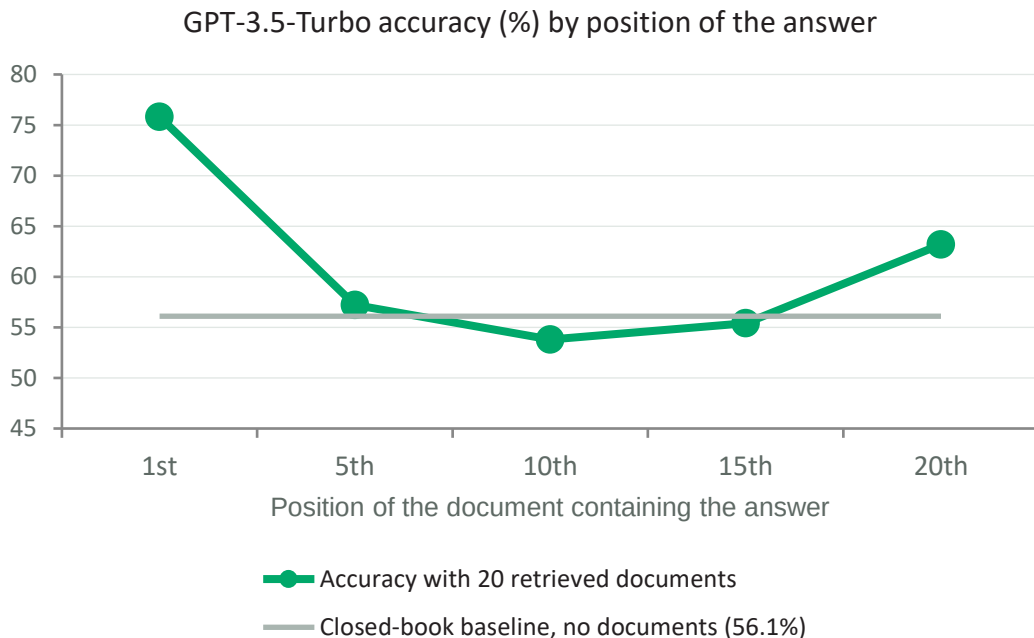
trainable parameters for RAG — vs 11B for T5-11B, which it outperforms.

42.7% vs 7.1%

Jeopardy generation: share of human judgments rating RAG vs BART as more factual.

Source: Lewis et al., NeurIPS 2020 (Tables 1 & 4). SSM = salient span masking pre-training.

Lost in the Middle: position matters



-22 pts

from 75.8% (answer first) to 53.8% (answer in the middle).

Below baseline

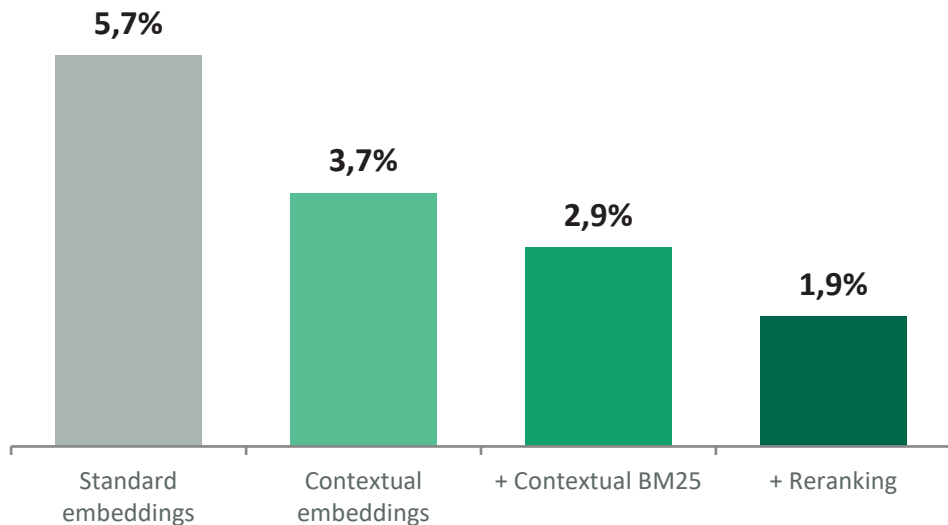
Mid-context, accuracy is worse than giving the model no documents at all (56.1%).

For RAG: rerank so the best evidence comes first, and send fewer, better chunks.

Source: Liu et al., "Lost in the Middle", TACL 2024 (Table 6). Multi-document QA on NaturalQuestions-Open.

Contextual Retrieval: restoring lost context

Top-20 retrieval failure rate (%) — lower is better



Source: Anthropic, “Introducing Contextual Retrieval”, September 2024.

The problem

Once split, a chunk loses who and when it refers to:

Before: “Revenue grew by 3% over the previous quarter.”

After: “[ACME Corp, Q2 filing] Revenue grew by 3%...”

The fix

An LLM writes a short context (≈50–100 tokens) situating each chunk within its document, prepended before embedding and BM25 indexing.

–49% failures · **–67%** with reranking

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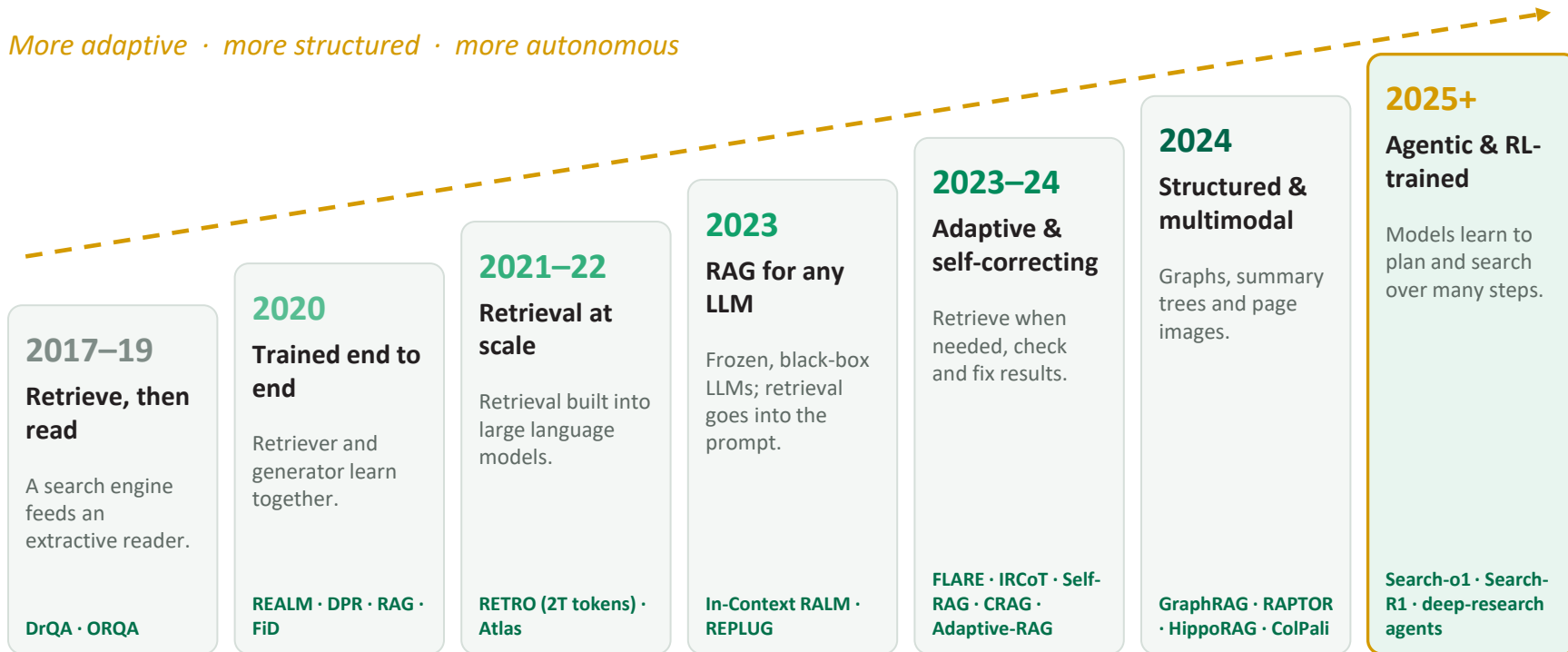
Evolution of RAG

How has the idea of RAG matured since 2017?



How the RAG concept has progressed

More adaptive · more structured · more autonomous



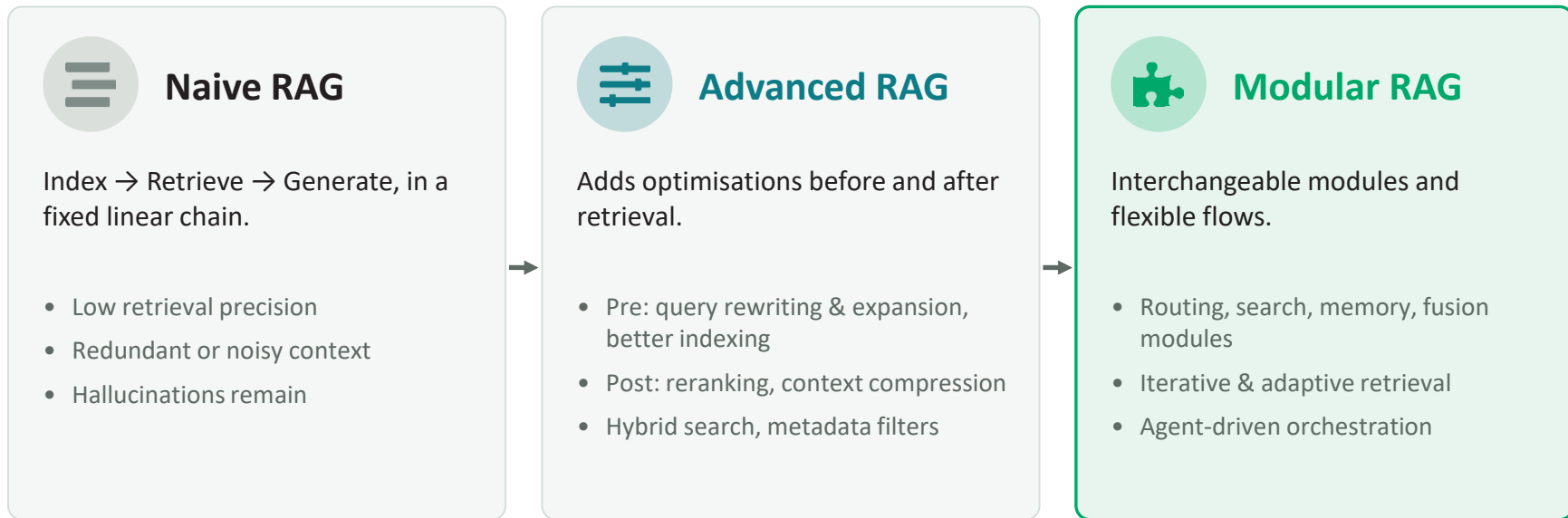
Five dimensions of progress

Dimension	Early RAG (2020)	Established practice	Frontier
When to retrieve	Once, before generating	Several times, interleaved with reasoning (IRCoT, FLARE)	The model decides if and when (Self-RAG, Adaptive-RAG)
What is retrieved	Fixed-size text passages	Hybrid, reranked, contextualised chunks	Knowledge graphs, summary trees, page images (GraphRAG, RAPTOR, ColPali)
How search is learned	Retriever trained jointly with a small generator	Frozen LLM; retriever off-the-shelf or tuned from LLM feedback (REPLUG)	LLM trained by reinforcement learning to search (Search-R1)
Who controls the flow	Fixed linear pipeline	Modular pipelines with routing	Agents that plan multi-step research
Quality control	Ranking only	Reranking, citations, automated evaluation (RAGAS)	Self-critique and corrective retrieval (Self-RAG, CRAG)

Direction of progress



From Naive to Advanced to Modular RAG

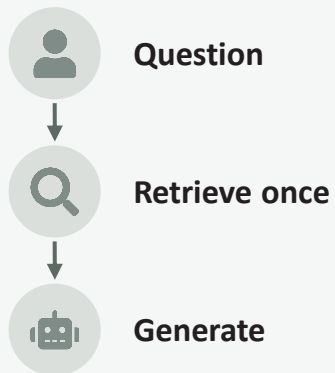


Additive, not a replacement: each paradigm builds on the components of the previous one.

Framework from Gao et al., "Retrieval-Augmented Generation for Large Language Models: A Survey", 2023.

From a one-shot pipeline to a reasoning loop

BEFORE · ONE SHOT



If the first search misses, the answer fails.

IRCoT *Trivedi et al., 2023*

Interleaves retrieval with each chain-of-thought step.

NOW · AGENTIC LOOP



FLARE *Jiang et al., 2023*

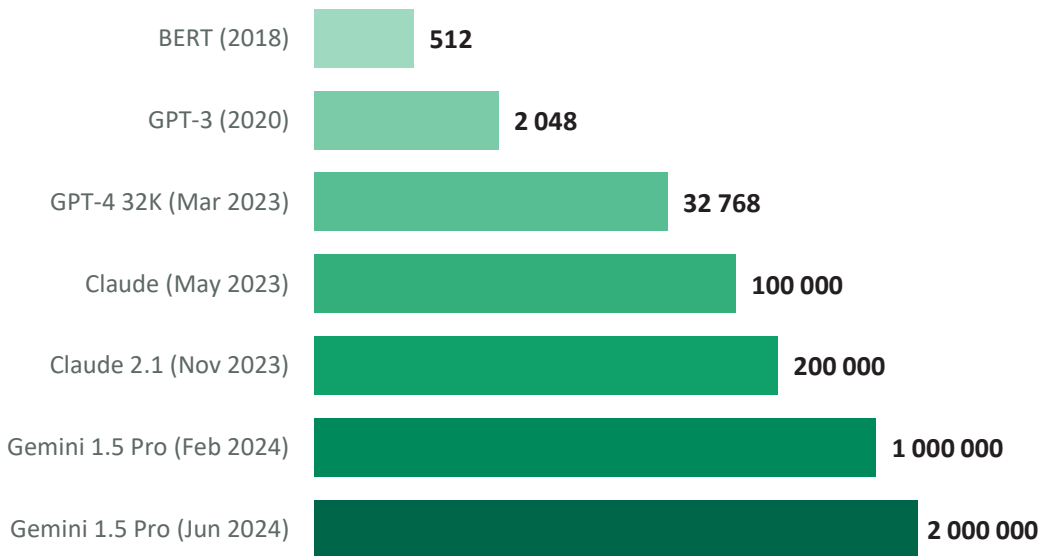
Retrieves when the next sentence contains low-confidence tokens.

Search-R1 *Jin et al., 2025*

LLM trained with RL to issue multiple searches while reasoning.

Bigger context windows changed RAG's role

Maximum context window (tokens, log scale)



Figures as announced by each provider.

Retrieval-Augmented Generation (RAG)

Then: RAG was a workaround — models could read only a few pages at once.

Evidence: long-context models can beat RAG on average, but RAG costs far less. Self-Route (Li et al., 2024) tries RAG first and falls back to full context only when needed.

Now: RAG is the layer for cost, freshness, access control and precision.

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Advanced techniques

Which techniques push answer quality further?



Advanced RAG techniques in practice



Gao et al., 2022

HyDE

LLM writes a hypothetical answer; its embedding is used to search.



Query transformation

Multi-query rewriting

Rewrites or splits a question into sub-queries, then merges results.



Parent-document

Small-to-big retrieval

Matches on small chunks, returns the larger parent section.



Asai et al., 2023

Self-RAG

Model decides when to retrieve and critiques outputs via reflection tokens.



Yan et al., 2024

Corrective RAG (CRAG)

An evaluator grades retrieved docs and triggers corrective actions such as web search.



Microsoft, 2024

GraphRAG

Builds a knowledge graph + community summaries to answer global questions.



Sarathi et al., 2024

RAPTOR

Recursive tree of clustered summaries for multi-level retrieval.



Agents + tools

Agentic RAG

An agent plans, calls retrieval tools iteratively and verifies before answering.



Query & indexing



Research methods



Agentic

RAG vs fine-tuning vs long context

	RAG	Fine-tuning	Long context
Updating knowledge	Re-index documents	Retrain the model	Change the prompt
Source citations	Native (retrieved chunks)	Not available	Possible
Knowledge volume	Millions of documents	Absorbed into weights, imprecise	Bounded by the context window
Cost per query	Low — only relevant chunks	Low	High for large contexts
Access control	Filter by user rights at retrieval	Very hard	Per request
Best for	Private & changing facts	Style & specialised behaviour	Small, self-contained corpora



Complementary, not exclusive: fine-tune for behaviour, RAG for knowledge, long context for small corpora.

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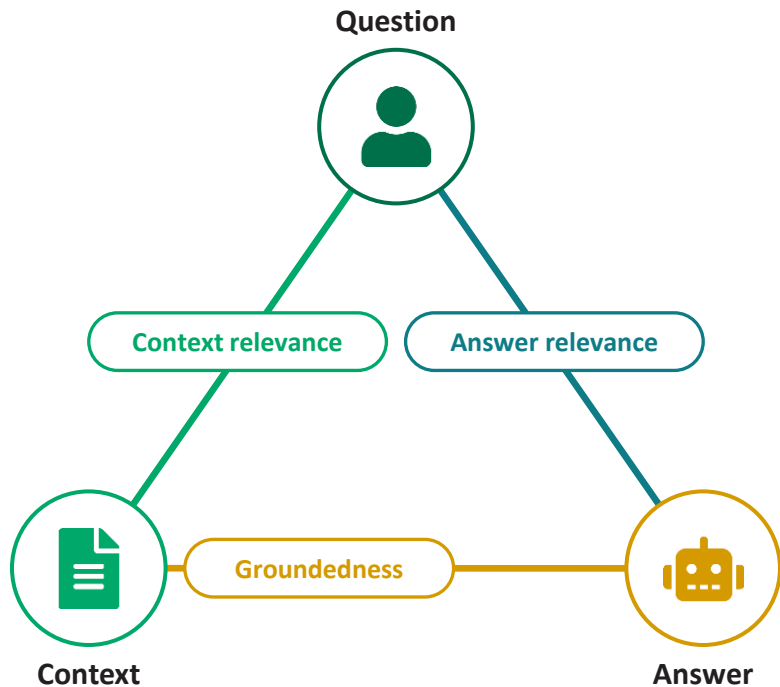
06

Production

How do you run RAG reliably and safely?



Measure retrieval and generation separately



The “RAG triad” (TruLens)

RAGAS metrics (Es et al., 2023)

Faithfulness: Are the answer's claims supported by the context?

Answer relevancy: Does the answer address the question?

Context precision: Are relevant chunks ranked at the top?

Context recall: Was all needed information retrieved?

Retrieval metrics

Recall@k · Precision@k · MRR · nDCG

Tools: RAGAS · TruLens · DeepEval · Arize Phoenix

Securing a RAG system



Indirect prompt injection

Hidden instructions inside ingested documents are read by the LLM as if they were commands.



Knowledge poisoning

PoisonedRAG (Zou et al., 2024): 5 crafted texts per target question reached ~90% attack success in a corpus of millions.



Permission leakage

Vectors do not inherit source-system permissions; users or tenants may retrieve what they should not see.



Embedding inversion

Research shows much of the original text can be reconstructed from its embedding (Morris et al., 2023).



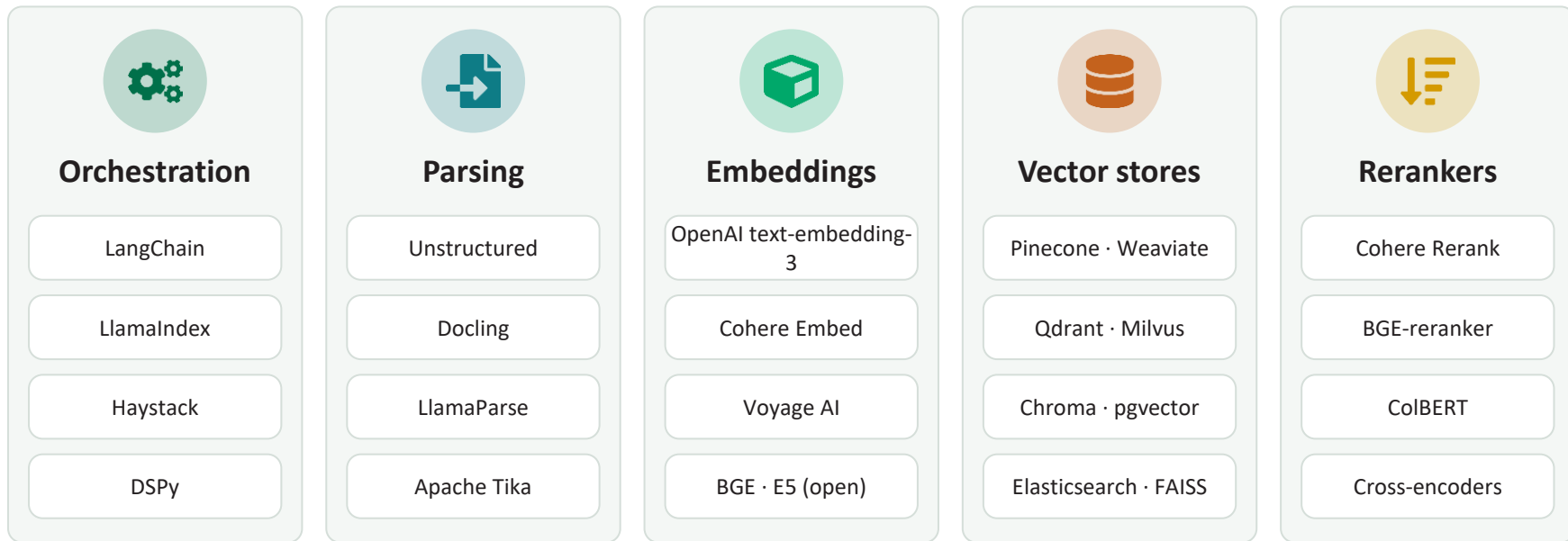
Mitigations

- Validate and sanitise documents at ingestion
- Enforce access control at retrieval; partition tenants
- Treat retrieved text as data, never as instructions
- Restrict what tools and actions the LLM can trigger
- Encrypt vector stores; log and monitor retrievals



OWASP Top 10 for LLM Applications 2025 now lists “LLM08: Vector and Embedding Weaknesses”.

The RAG technology stack



Tip: compare embedding models on the MTEB benchmark, then validate on your own data and languages (e.g. French, Arabic).

Failure points and best practices



7 failure points

Barnett et al., 2024

- 1 Missing content — the answer is not in the corpus
- 2 Missed top-ranked documents
- 3 Not in context — dropped during consolidation
- 4 Not extracted — present but ignored by the LLM
- 5 Wrong format
- 6 Incorrect specificity
- 7 Incomplete answer



Best practices

What consistently works

- ✓ Build an evaluation set before optimising
- ✓ Clean parsing & structure-aware chunking
- ✓ Hybrid search + reranking as a default
- ✓ Metadata filters and permission checks
- ✓ Grounding prompt with citations & abstention
- ✓ Monitor queries, refresh the index, iterate

KEY TAKEAWAYS

Retrieval quality is the ceiling of answer quality.



Grounded

RAG gives LLMs current, private and citable knowledge without retraining.



Retrieval first

Chunking, hybrid search and reranking drive most of the gains.



Measured

Evaluate retrieval and generation separately, on your own data.



Evolving

From fixed pipelines to adaptive, agentic retrieval that decides when and what to search.

Start simple · *measure* · **iterate on retrieval.**



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